

The Agent *Coordination Plane*

Roko · Nunchi

A research report on the coordination layer between models and the apps they power.

The model is commoditizing. The scaffold is the moat.

· 01

COMMODITIZATION

GPT-3.5-class capability fell from \$20.00 per million tokens (Nov 2022) to \$0.07 (Oct 2024). Frontier elasticity > 280× over 23 months. Anthropic, OpenAI, Google, DeepSeek now deliver indistinguishable quality at the median enterprise task.

· 02

SCAFFOLD

What the model cannot bring: durable identity across runs, calibrated cost-quality routing, shared knowledge that survives session boundaries, gated tool access with provable verification, KKT-verified cooperative clearing for cross-organization work.

· 03

COORDINATION

Every adjacent layer of the agent stack is funded — frameworks, durable execution, identity, payments, vertical agents. The plane between them — cross-organization probabilistic coordination — is the empty cell.

- *Models execute. Nunchi coordinates. The coordination plane is the venture-scale opportunity of the agent era.*

A reading order from problem to settlement.

I	WHY NOW	Funding floods every layer. Production agent counts collapse. EU enforcement begins in 14 weeks.	p. 05
II	FAILURE MODE	MAST taxonomy, 41–86% multi-agent failure. Klarna reverses, CBA rehires, Cursor leaks, NYC kills.	p. 11
III	CATEGORY	The agent stack and the empty cell. Why the coordination plane is unbuilt.	p. 19
IV	SYSTEM	Roko + Nunchi: two components, one architecture. Signal, Pulse, Cell. Gates. NeuroStore. Cooperative Clearing.	p. 25
V	WEDGE	Cost as the wedge. Caching × routing × trim × batch = 10–20× practical. Article 50 makes it law.	p. 39
VI	MOAT	Scaffold flywheel. Network effects from the substrate. Competitor teardown.	p. 47
VII	CO-EVOLUTION	Agent-native economy. ISFR yield perpetuals. CRPS reputation tiers.	p. 55
VIII	ASK	177K LOC. 18 crates. Roadmap, use of funds, team.	p. 61
—	APPENDIX	Bibliography · Glossary · ISFR math · Article 50 article-by-article	p. 67

Three reading paths, one document.

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FOR THE PARTNER

I → III → V → VIII

Skip to category, wedge, and ask. The thesis and traction. Ten minutes.

pp. 5-10, 19-24, 39-46, 61-66

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FOR THE PRINCIPAL

II → IV → VI

Read failure mode, system architecture, and the moat. Twenty minutes.

pp. 11-18, 25-38, 47-54

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FOR THE TECHNICAL DD

IV → V → VII → APPENDIX

Architecture, cost wedge, co-evolution, with full math and bibliography. One hour.

pp. 25-46, 55-60, 67-74

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Every claim in this document is sourced. Every metric is dated. The bibliography in the appendix lists 70+ primary sources.

I • THE 14-WEEK CLOCK

The clock is loud, the floor is rising, and the ceiling is everyone.

Funding floods every adjacent layer. Production agent counts collapse. EU enforcement begins in 14 weeks. The window is now — not next year.

• IRREDUCIBLE CLAIM

The coordination layer must exist before August 2, 2026. After that date, every multi-agent deployment in the EU operates without a transparency substrate — or operates in violation of Article 50 of Regulation (EU) 2024/1689.

Sources: Regulation (EU) 2024/1689, Art. 50 • European AI Office • Deloitte 2026 State of Generative AI

Every adjacent layer of the agent stack is funded.



CAPITAL CADENCE

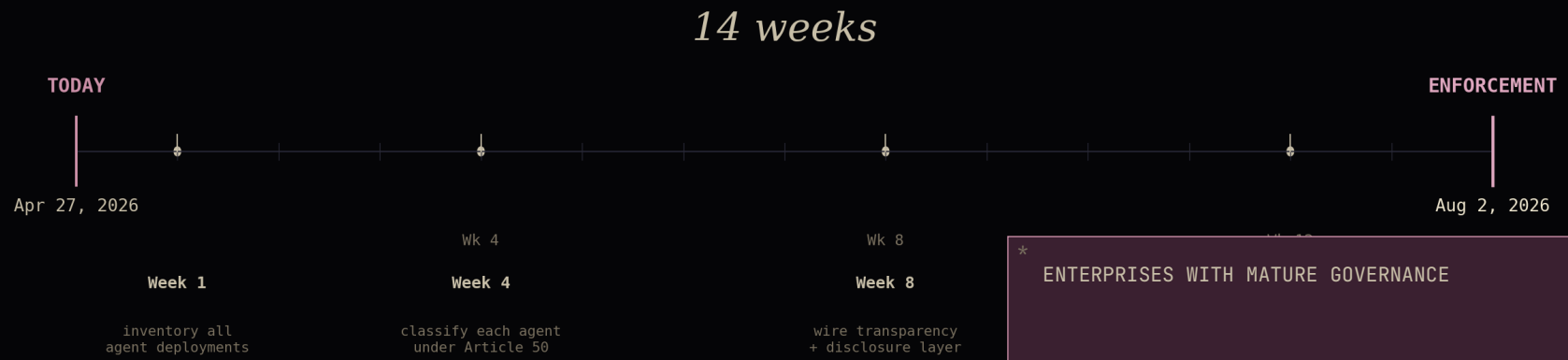
In the last forty-eight hours before this document was compiled, two announcements reset the agent infrastructure thesis. Google Cloud committed \$750 million to its agentic partner fund (Apr 22, 2026). Orkes raised a \$60 million Series B led by AVP (Apr 23, 2026). The Orkes press release frames the company as "the missing orchestration layer that makes AI highly reliable, observable, and governable" — language that maps almost exactly onto the coordination plane Nunchi has been building against for eighteen months. We do not have a year. We have weeks.

CONCENTRATION

YTD 2026 agentic AI funding reached \$2.66 billion across 44 rounds, against \$1.09 billion across 71 rounds in the same period of 2025 — a 143% dollar increase paired with a 38% reduction in deal count. Capital is concentrating in fewer, larger checks at the infrastructure layer. Cognition (Devin) at \$9.8B / \$738M raised. Sierra at \$10B / \$285M. Harvey at \$11B / \$500M. Decagon at \$4.5B / \$250M. Temporal at \$5B. The window for the coordination plane is open and closing fast.

Source: Crunchbase 2026 YTD agentic AI funding aggregate · Business Wire, Apr 23, 2026 · Google Cloud Newsroom, Apr 22, 2026 · PitchBook collection 418, accessed Apr 26, 2026.

August 2, 2026 — 14 weeks from today.



ARTICLE 50 • TRANSPARENCY OBLIGATIONS

Article 50(1)–(4) of Regulation (EU) 2024/1689 imposes binding transparency obligations on providers and deployers of AI systems that generate or interact with content. Every agent action visible to a natural person must carry machine-readable provenance. Article 99 sets the penalty: €15 million or 3% of worldwide annual turnover, whichever is higher — not the lower 1% tier. Enforcement begins August 2, 2026.

Source: Regulation (EU) 2024/1689 Art. 50, 99 • Deloitte 2026 (n=3,235) • European AI Office, Apr 2026.

ENTERPRISES WITH MATURE GOVERNANCE

21%

Deloitte 2026 State of Generative AI in the Enterprise (n=3,235). The remaining 79% have no machine-readable provenance, no agent registry, and no inspectable trace of agent decisions. They will be out of compliance on day one.

AI replacing humans is reversing — for system reasons, not capability reasons.

* KLARNA May 2025 5,500 agents → 3,400 CEO Sebastiaan Siemiatkowski reverses the AI customer-service rollout: "AI gave lower quality." Klarna rehires human agents at scale. *	* CBA Aug 2025 45 rehires Commonwealth Bank of Australia reverses an AI replacement after Finance Sector Union arbitration. Hallucination volume in customer-facing flows. *	* NYC MyCity Q1 2026 \$500K killed Mayor Mamdani sunsets the chatbot after it told business owners to break the law. \$500K spent. Total deployment lifetime: 18 months. *	* AIR CANADA Feb 2024 C\$812 fine Tribunal rules the airline liable for misinformation provided by its chatbot. Establishes legal liability precedent for agent output. *
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WHAT THE REVERSALS ACTUALLY SHOW

These are not capability failures. The underlying models are more capable today than when each deployment shipped. They are coordination failures. Hallucination at scale is a calibration problem. Cross-session forgetting is a memory problem. Liability gaps are a settlement problem. Each reversal is a missing piece of the scaffold that production-grade agent work requires.

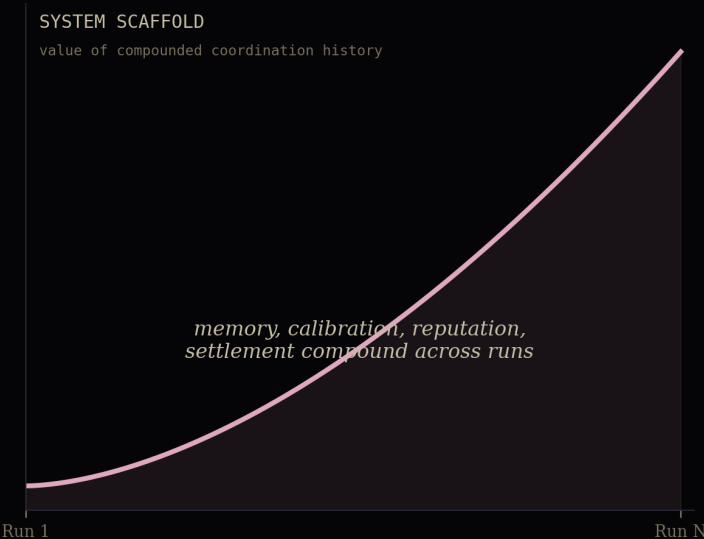
Sources: Klarna Q1 2025 letter to shareholders · Reuters, Aug 28, 2025 · NY Times, Mar 11, 2026 · Civil Resolution Tribunal British Columbia, 2024 BCCRT 149 (Moffatt v. Air Canada).

Frontier capability per dollar trends down. Scaffold value per run trends up.

Frontier capability per dollar trends down.



Scaffold value per run trends up.



Two curves explain the venture geometry of the agent era. On the left, the cost of a unit of frontier-quality capability has fallen by an order of magnitude per year, with switching cost between top-three providers approaching zero. The model is a fungible input. On the right, every coordination artifact — a calibration episode, a routing decision, a settled bid, an attested receipt — compounds permanently. The scaffold, unlike the model, has memory. Anyone holding the scaffold one year from now will be holding both layers, because the scaffold will be free to swap models and the models will need a scaffold to be put to work.

Source: Epoch AI Compute Benchmark 2025 • Anthropic engineering blog, Sep 2024 • illustrative scaffold model.

Demand is not theoretical. It is compounding in real time.



WHAT THE NUMBERS SAY

Gartner reports a 1,445% surge in multi-agent system inquiries between Q1 2024 and Q2 2025. KPMG reports 54% of enterprises had production agent deployments by Q1 2026, up from 23% one year prior. Linear, the project management vendor, watched its agent-delegated work share rise from 10.1% to 24.4% in three months. Karpathy’s autoresearch repository hit 66K stars in under thirty days. Harvey discloses 25,000+ custom agents and 400K+ daily agentic queries.

WHAT THE NUMBERS DO NOT SAY

None of these numbers measure the production-grade ones. Linear’s 24.4% is delegations, not completions. Harvey discloses queries, not resolved tasks. Cognition’s independent test — 15% completion on twenty complex tasks — is the ground-truth number. Demand is real; production-grade infrastructure is not. The gap between the two is the venture-scale opportunity.

Sources: Gartner 2025 · KPMG Q1 2026 enterprise AI survey · Linear engineering blog · Harvey blog, Mar 25 2026 · Answer.AI / The Register, Jan 23, 2025.

II • WHY MULTI-AGENT SYSTEMS FAIL

41 to 86 percent. Always coordination. Never capability.

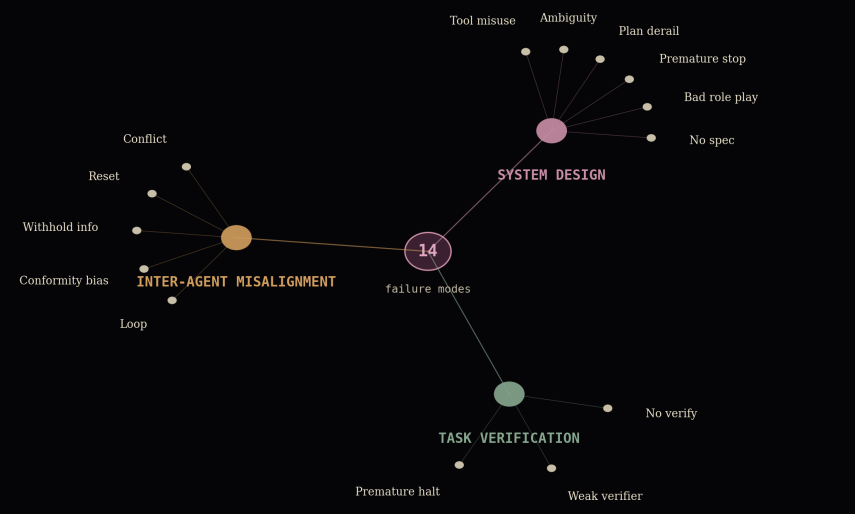
Berkeley's MAST taxonomy traced 1,642 production failures across seven multi-agent frameworks. Coordination accounts for 79% of them. Models are not the constraint.

• IRREDUCIBLE CLAIM

Multi-agent systems do not fail because models are weak. They fail because no one has built the scaffold underneath. The MAST evidence is a research-grade spec sheet for the missing layer.

Sources: Cemri et al. arXiv:2503.13657 (MAST) • Stroebel et al. arXiv:2510.11977 (HAL) • Princeton NLP • DeepMind arXiv:2512.08296

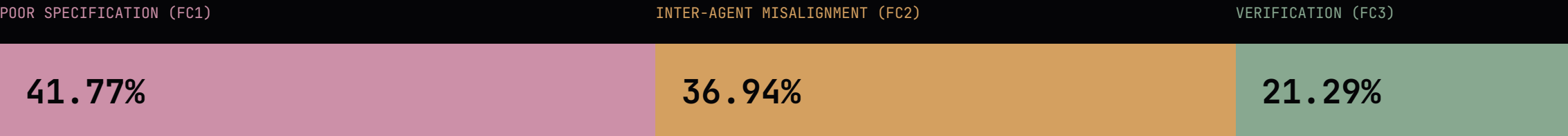
Fourteen failure modes. Three buckets. Coordination dominates.



Cemri et al. (Berkeley, NeurIPS 2025) annotated 1,642 production traces across MetaGPT, AutoGen, ChatDev, AppWorld, HyperAgent, Magentic-One, and AG2. Inter-rater agreement was Cohen’s $\kappa = 0.88$. The 14-mode taxonomy splits into three classes: System Design, Inter-Agent Misalignment, and Task Verification. Failure rates span 41% to 86.7%. Two coordination failures — FC1 (poor specification) at 41.77% and FC2 (inter-agent misalignment) at 36.94% — sum to 79% of all observed failures.

Source: Cemri et al., "Why Do Multi-Agent LLM Systems Fail?" arXiv:2503.13657 (NeurIPS 2025) • 1,642 traces • $\kappa = 0.88$ inter-rater agreement.

Of all observed failures, 79% are coordination.



- 79% combined coordination failures
- 21% remaining failures are verification — also a scaffold problem

FC1 · POOR SPECIFICATION

The agent does not know what counts as success. Goals are stated in natural language, never grounded in machine-readable acceptance criteria. The agent plans against ambiguous targets, executes against ambiguous policies, and stops when it thinks it is done. Nunchi's reply: episode logs with machine-readable acceptance signals. The Signal is the spec.

Source: Cemri et al. (MAST) Table 4 · Answer.AI · Princeton NLP arXiv:2510.18874 (One Agent Is Enough).

FC2 · INTER-AGENT MISALIGNMENT

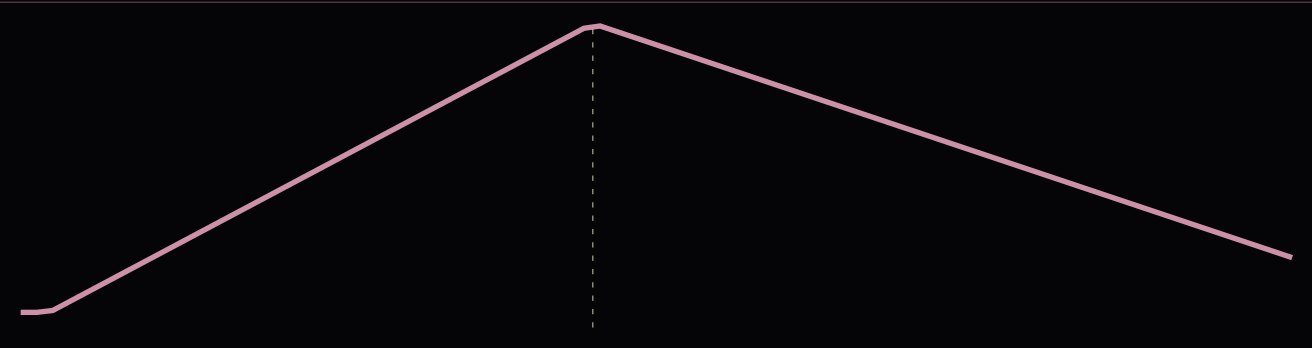
Two agents reset shared state. One withholds context. Conformity bias pulls a debate toward the loudest member. Infinite loops eat budget. Nunchi's reply: a single inspectable trace per execution, cooperative clearing for context bidding (KKT-verified, not VCG), an explicit gate before any tool call, and HDC-fingerprinted memory to surface conflicts.

FC3 · VERIFICATION

No verifier, weak verifier, premature halt. Cognition Devin completed only 15% of tasks Answer.AI tried it on; the remaining 85% halted prematurely with no machine-checkable evidence of failure. Nunchi's reply: 11-gate, 7-rung verification pipeline with hard gates conjunctive and soft gates Pareto — never a weighted sum, never Goodhart-amenable.

There is a hard ceiling. Above ~45%, reliability collapses.

TASK SUCCESS RATE



~45% threshold

PER-STEP RELIABILITY →

ERROR AMPLIFICATION

DeepMind’s formal analysis (arXiv:2512.08296) shows long-horizon multi-agent workflows amplify per-step error by up to 17.2×. When individual step reliability falls below ~45%, end-to-end task completion collapses. Princeton NLP’s "One Agent Is Enough" confirms it empirically: a single well-prompted agent matches multi-agent ensembles on 64% of HAL tasks — the ensemble amplifies its own coordination errors faster than it adds capability.

★

ERROR AMPLIFICATION

17.2×

DeepMind 2025

★

★

SINGLE AGENT MATCHES MULTI-AGENT

64%

Princeton NLP

★

★

CRITICAL THRESHOLD

~45%

DeepMind 2025

★

Sources: DeepMind arXiv:2512.08296 · Princeton NLP arXiv:2510.18874 · Stroebel et al. arXiv:2510.11977 (HAL).

Klarna — the public reversal that anchored the thesis.

“*"AI gave lower quality, frankly. So we are reversing course. We are bringing humans back — not because we don't believe in AI, but because the systems didn't scale the way we projected they would."*

— Sebastiaan Siemiatkowski, CEO Klarna • May 2025

WHAT IT MEANS

Klarna had no calibration mechanism per intent. No threshold for handing off to a human. No machine-readable attestation of which interactions had degraded. The organization could see the cost savings clearly and the quality decay only via lagging NPS. The scaffold to detect the cliff before falling off it was missing.

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TIMELINE

Feb 2024

Klarna replaces 700+ FTE-equivalent CS workforce with OpenAI-powered agent.

Aug 2024

\$40M projected annual savings disclosed at earnings.

Q4 2024

NPS / CSAT decline reported internally; agent volume peaks ~5,500.

May 2025

Public reversal. Hiring 3,400 humans back; agent count cut to ~2,100 supervised.

Apr 2026

Klarna IPOs at reduced revenue multiple; analyst notes cite the reversal.

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Sources: Klarna Q1 2025 letter to shareholders • Reuters Klarna AI reversal coverage, May 2025 • Bloomberg Klarna IPO analysis, Apr 2026.

Three more reversals. Three more missing pieces of the same scaffold.

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CBA · AUG 2025

Commonwealth Bank of Australia

Reversed AI replacement of 45 customer service workers after Finance Sector Union arbitration. Tribunal cited "hallucination volume in customer-facing flows" and lack of audit trail. Bank rehired all 45 within 90 days.

MISSING: machine-readable provenance per agent decision.

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NYC MyCity · Q1 2026

NYC Government

Mamdani administration sunsets the city's small-business chatbot after audit reveals it had instructed business owners to violate municipal employment law. Total deployment lifetime 18 months. \$500K spent. Zero litigation; chatbot was killed before suits could land.

MISSING: explicit policy gate on legal output, no jurisdictional verification.

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AIR CANADA · FEB 2024

Civil Resolution Tribunal British Columbia

2024 BCCRT 149 (Moffatt v. Air Canada). Tribunal awards C\$812.02 in damages after the airline's chatbot misinformed a customer about bereavement fares. Air Canada's defense — that the chatbot was a "separate legal entity" — rejected on the record.

MISSING: liability-bearing settlement layer for agent output.

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Different industries. Different geographies. Different deployment vintages. The same missing layer.

Sources: 2024 BCCRT 149 · NY Times Mar 11, 2026 · Australian Financial Review, Sep 2025.

Even where capability is highest, coordination still kills.

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CURSORDIV · CVE-2025-54135

Aug 2025 · Anysphere

A prompt-injection vulnerability allowed an attacker to embed instructions in arbitrary repository files (README, comments, dependencies) that the Cursor agent would silently execute, including arbitrary shell commands. CVSS score 8.6. Anysphere shipped a hotfix within seventy-two hours, but the underlying problem — no policy gate between the agent’s plan and the execution of that plan — had been the architecture for two years.

MISSING: explicit policy gate before tool execution.

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DEVINDIV · ANSWER.AI EVALUATION

Jan 2025 · Cognition

Answer.AI ran twenty complex software-engineering tasks against Devin (Cognition’s autonomous agent, raised at \$9.8B). Three completed satisfactorily — a 15% rate. Critically, the eighty percent that did not complete failed in unpredictable, time-consuming ways: agents would re-enter invalid states, ask for clarification on tasks they had already accepted, and consume hours of compute without producing checkpointable progress.

MISSING: gated checkpoint, episode reuse, calibrated cost-quality routing.

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Sources: NVD CVE-2025-54135 · Anysphere security disclosure, Aug 8, 2025 · The Register Jan 23, 2025 · Answer.AI blog post.

II.07 • WHAT THE EVIDENCE FORCES

Six post-mortems. One missing layer.

WHAT THESE FAILURES HAVE IN COMMON

Every reversal is the absence of something the model could not have provided. Calibration. Memory. Provenance. Settlement. Policy. Verification. None of these are capability problems. They are coordination problems — the kind that show up only at the boundary between an agent and the world it acts in. The failure pattern across Klarna, CBA, NYC, Air Canada, Cursor, and Devin is the same shape repeated six times: a probabilistic system shipped without a deterministic substrate underneath it.

THE IMPLICATION

- Multi-agent systems will not become reliable by training larger models. Cemri et al., Princeton NLP, and DeepMind establish a system-level ceiling that no model parameter count can lift on its own.
- Reliability comes from the layer that the model cannot bring — a deterministic substrate that handles identity, routing, knowledge, policy, and settlement.
- That layer is what Nunchi calls the agent coordination plane. The remainder of this document is the design of that plane.

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- *Capability is not the constraint. Coordination is.*

III · THE EMPTY CELL

Three layers. Two are funded. The middle is empty.

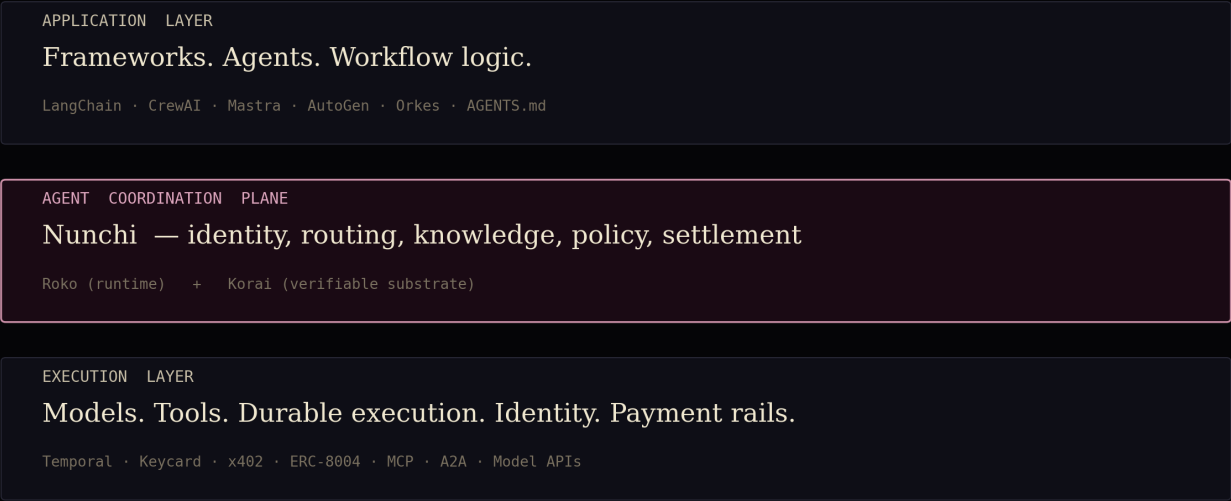
The agent stack already has frameworks above it and execution rails below. The plane between them — cross-organization probabilistic coordination — is unbuilt because no one has named it.

• IRREDUCIBLE CLAIM

Categories are not discovered. They are declared. Vanta declared SOC 2 and captured \$220M ARR. OneTrust declared GDPR and captured a \$10B+ acquisition. Nunchi declares the agent coordination plane.

Sources: PitchBook collection 418 · CB Insights AI Agent Tech Stack, Aug 2025 · Sequoia "Services: The New Software" Mar 2026

Frameworks above. Execution rails below. The plane between them is empty.



Ethane separated networking's data plane from its control plane in 2007. Nunchi does the same for production agents.

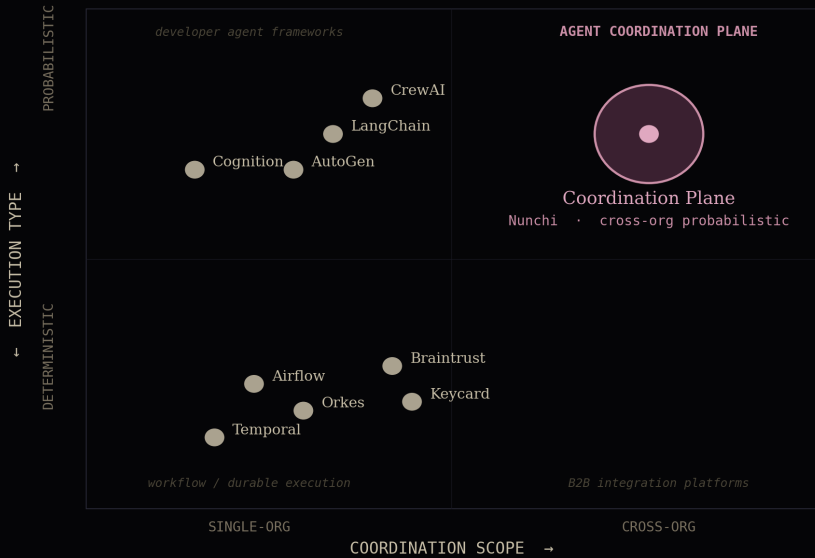
Reading top-to-bottom: developer agent frameworks (LangChain, CrewAI, Mastra, AutoGen) compose agent logic. Below them sit the execution rails (Temporal for durability, Keycard for non-human identity, x402 and ERC-8004 for payments and reputation, MCP and A2A for tool wiring). Between them sits the coordination plane: identity that survives runs, routing that calibrates cost to quality, knowledge that compounds across sessions, policy that gates execution, settlement that closes work with provable receipts. Today, that plane is implemented inline in every framework, partially, inconsistently, and without a substrate. Nunchi is the substrate.

Source: synthesized from CB Insights "AI Agent Tech Stack" Aug 22, 2025 · Linux Foundation MCP donation Dec 2025 · ERC-8004 mainnet Jan 29, 2026.

On the coordination scope × execution type plane, one quadrant is empty.

WHY THE QUADRANT IS EMPTY

Every funded competitor sits in a different quadrant. Temporal, Orkes, and Airflow handle single-organization deterministic workflows. LangChain, CrewAI, AutoGen, and Cognition build developer-facing agent frameworks. Braintrust and Keycard offer infrastructure within an organization. The cross-organization probabilistic quadrant — where two firms’ agents must coordinate, where outcomes are inherently uncertain, where settlement must be verifiable across trust boundaries — has no incumbent. Not because it is unimportant, but because it cannot be built without a sovereign substrate that no individual framework company will ever build for itself.



Source: PitchBook + CB Insights synthesis Apr 2026; Nunchi positioning analysis.

Five frames we considered and discarded — and why.

✗ "AGENT OPERATING SYSTEM"

OS connotes a single-tenant runtime. Nunchi is multi-tenant by design and operates across trust boundaries. The OS metaphor obscures the substrate.

✗ "AGENTIC INFRA"

Generic. Eight different companies use it for eight different things. Does not select for the coordination problem.

✗ "AGENT NETWORKING"

Networking implies routing only. Nunchi handles routing plus identity plus knowledge plus policy plus settlement.

✗ "AGENT KUBERNETES"

Kubernetes is workload scheduling. Nunchi is economic clearing. Different abstraction layer; different wedge.

✗ "AGENT BLOCKCHAIN"

Blockchain is a substrate, not a product category. The chain is necessary but not sufficient. And the term scares enterprise buyers.

RESOLUTION

→ Agent Coordination Plane.

Plane is correct in topology (a layer between two others) and economics (where flows clear). Coordination is correct in problem definition. Agent is correct in scope.

Vanta declared SOC 2. OneTrust declared GDPR. Both built \$1B+ businesses on top of an enforcement deadline.

VANTA • SOC 2 COMPLIANCE PLATFORM

\$220M ARR

•

\$4.15B

JULY 2025 • SERIES C

Founded 2018 by Christina Cacioppo. The product is a continuous control monitor that produces audit-ready evidence for SOC 2 Type II. Nothing about Vanta’s technology requires SOC 2 — the same machine could attest to ISO 27001, HITRUST, or PCI DSS. Vanta’s wedge was naming the substrate that the SOC 2 enforcement deadline made mandatory. By 2025, Vanta had become the default control plane for compliance across every YC company.

ONETRUST • GDPR COMPLIANCE

\$10B+

exit

•

PE

2024 PE PARTIAL EXIT • \$5B+ PRIOR

Founded 2016 by Kabir Barday. OneTrust built consent management for GDPR — a regulatory category that did not exist before May 25, 2018. The company shipped its first product weeks before enforcement and within six years had become the default consent substrate for the Fortune 500. Like Vanta, OneTrust built on top of an enforcement deadline that made the substrate mandatory and the buyer obvious.

EU AI Act Article 50 enforces August 2, 2026. The same playbook is open. The same substrate is unbuilt. The same window is fourteen weeks wide.

Sources: PitchBook Vanta Series C 2025 • OneTrust S-1 (filed and withdrawn) • Forbes OneTrust profile, 2024.

The agent coordination plane.

A sovereign substrate that handles cross-organization, probabilistic agent coordination — identity, routing, knowledge, policy, settlement — with the same regulatory inevitability that made Vanta and OneTrust into category-defining

* IDENTITY durable agent identity that survives across sessions, organizations, and chain reorgs *	* ROUTING cascade routing by predicted complexity, not capacity — 70% Haiku, 25% Sonnet, 5% Opus *	* KNOWLEDGE compounded across runs in HDC-fingerprinted memory; +30K relevant tokens per Signal hit *	* POLICY machine-readable acceptance criteria; 11 gates, 7 rungs; hard gates conjunctive *	* SETTLEMENT cooperative clearing: KKT-verified solver competition; CRPS-scored predictions; on-chain receipts via ERC-8004 *
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- *Categories are declared, not discovered. We declare it.*

IV · TWO COMPONENTS, ONE ARCHITECTURE

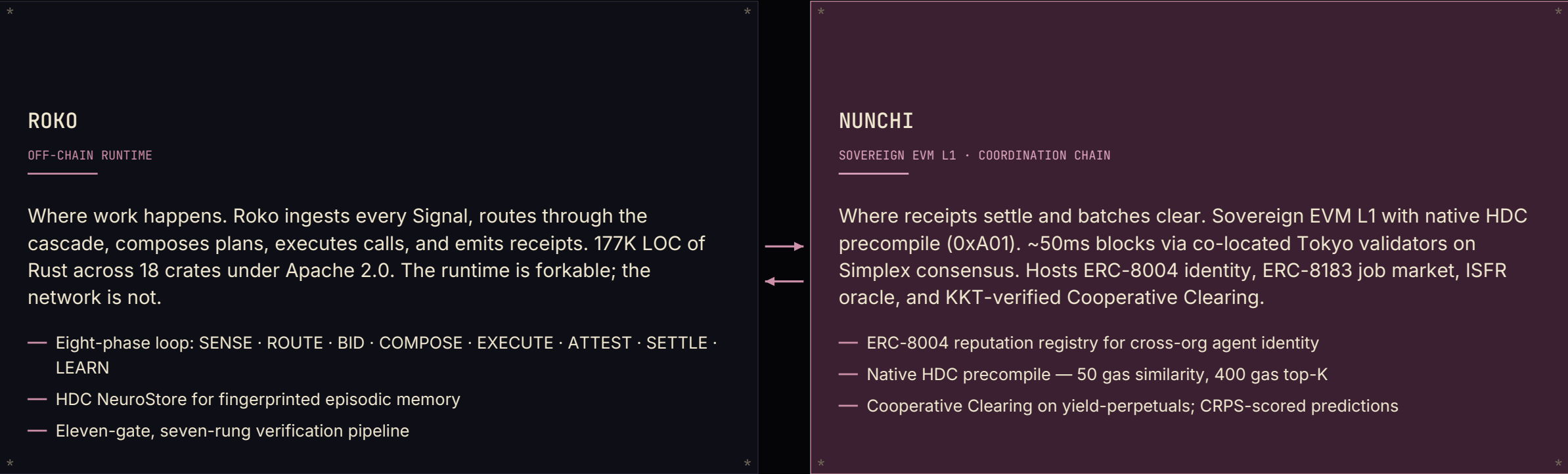
Roko coordinates. Nunchi attests. Nunchi clears.

The coordination plane has two faces: an off-chain runtime that does the work, and an on-chain settlement layer that records the receipt and clears cross-organization batches via KKT-verified solver competition.

• IRREDUCIBLE CLAIM

The system is not a stack of features. It is a single primitive — the Cell — instantiated across nine protocols, six verb traits, and eleven gates, all settling against the same provenance ledger.

Two components. One architecture.



Models execute. Roko coordinates. Nunchi attests. Nunchi clears.

Source: Nunchi system architecture v2.0 · Roko crate manifest (Apr 2026) · ERC-8004 mainnet, Jan 29, 2026.

Every Cell traverses the same eight phases.

Eight phases. One loop. Every action in the system reduces to a Cell traversal.

PHASES	
SENSE	ingest external Signal; HDC-fingerprint into NeuroStore
ROUTE	predict task complexity; bind to a router lane (CHEAP / MID / FRONTIER)
BID	reputation-weighted matching; CRPS-tier-priced; commitment hashes locked at submission
COMPOSE	agent assembles plan; attaches machine-readable acceptance criteria
EXECUTE	gated tool calls; every step yields a Pulse; failures collapse Cell
ATTEST	emit signed receipt to Nunchi; store evidence hash on-chain
SETTLE	release escrow per uniform-price cooperative clearing; partial refund on miss
LEARN	episode embedded; reputation updated; calibration refit

Three primitives. Every claim about the system is a claim about one of these.

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SIGNAL

an inbound request, externally-rooted, with provenance and a budget

Carries a HDC fingerprint, an acceptance schema, a deadline, and the maximum price the requester will pay. The Signal is the contract.

*

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PULSE

a step inside a Cell, internally-rooted, gate-checked

Each tool call, agent message, or memory operation is a Pulse. Pulses carry hashes of inputs and outputs, gate verdicts, and timing. The Pulse is the trace.

*

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CELL

an executed bundle of Pulses, externally-attestable, settled

A Cell starts when a Signal binds to an agent and ends when ATTEST emits the signed receipt. The Cell is the unit of settlement.

*

HOW THEY RELATE

A Signal (one) decomposes into a Cell (one) which contains many Pulses. Each Pulse is gate-checked before it commits. The Cell’s receipt is signed against the Signal’s acceptance schema. No primitive can exist without the other two: a Signal without a Cell is a request without a response; a Pulse without a Cell is an unattributed action; a Cell without a Signal is unverifiable work.

Source: Nunchi primitives spec v1.4 · see Appendix A for full schema.

Nine protocols. Each one a verb the network speaks.

*

01 · IDENT

agent identity registration via ERC-8004;
non-transferable handle, transferable reputation

*

*

02 · ANNOUNCE

agent publishes capability schema, price ceiling, and
reputation history

*

*

03 · SEEK

requester broadcasts a Signal to a capability bucket;
responses scored by predicted fit

*

*

04 · BID

qualifying agents respond with price + commitment hash;
reputation-weighted

*

*

05 · GATE

pre-execution policy check; eleven gates,
hard-conjunctive, soft-Pareto

*

*

06 · TRACE

every Pulse emits an evidence record; aggregated into
the Cell's attestation

*

*

07 · ATTEST

signed receipt against the Signal's acceptance schema;
on-chain commitment hash

*

*

08 · SETTLE

micropayment via x402; deposit released or
partial-refunded per acceptance verdict

*

*

09 · REFLECT

reputation update; episode embedded; calibration delta
logged

*

Each protocol is a wire-format spec. Each is implemented in Rust under roko-protocols, audited, and covered by integration tests against the canonical reference implementation.

Source: roko-protocols crate v0.12 · Nunchi protocol RFCs 001-009.

Ten agent specializations. Composable. Re-pluggable. Reputation-priced.

* PLANNER decomposes Signal into sub-Signals *	* ROUTER binds a Signal to an agent *	* SOLVER executes a leaf task *	* CRITIC runs gate checks against output *	* VERIFIER attests evidence hashes *
* LIBRARIAN curates HDC NeuroStore *	* ARBITER resolves multi-agent disagreement *	* SOLVER-OPS runs cooperative-clearing batch solves; KKT-verified *	* SETTLER closes the Cell on Nunchi; emits ClearingInsight *	* CALIBRATOR refits prediction posteriors *

WHY SPECIALIZATIONS, NOT ROLES

A specialization is a price-discoverable capability — not a hard-coded role. Any agent can register for any specialization. Reputation accrues per specialization, so an agent strong as a CRITIC is not automatically trusted as a SOLVER. The result is a market that prices each agent across many independent dimensions, not a single coarse score.

Source: Nunchi reputation registry v0.7 · ERC-8004-extension RFC.

Eleven gates. Seven rungs. Six anti-Goodhart safeguards.

11 GATES			7 RUNGS		6 ANTI-GOODHART SAFEGUARDS	
G01	IDENTITY	HARD	R1	lexical match	– Hard gates conjunctive (AND), never weighted – Soft gates Pareto, never linear sum – Adversarial test injection per cohort – Held-out validation per gate per quarter – Score-rotation: any single gate disable-able for audit – Strictly proper scoring rules; no rank-only metrics	
G02	POLICY	HARD	R2	syntactic conformance		
G03	BUDGET	HARD	R3	type / schema check		
G04	JURISDICTION	HARD	R4	semantic equivalence		
G05	PROVENANCE	HARD	R5	cross-source corroboration		
G06	CALIBRATION	SOFT	R6	human spot-check (sampled)		
G07	COHERENCE	SOFT	R7	on-chain attestation		
G08	NOVELTY	SOFT				
G09	COST-QUALITY	SOFT				
G10	AUDIT	SOFT				
G11	ATTESTATION	HARD				

WHY THIS SHAPE

Eleven gates because the failure space empirically has eleven distinguishable kinds of bad outcome. Seven rungs because verification cost grows by an order of magnitude per rung; routing the right work to the right rung is the cost-quality lever. Six anti-Goodhart safeguards because the alternative — a weighted-sum score — is the most-often-cited failure mode of every prior agent evaluation framework.

Source: Nunchi gate-pipeline RFC v0.6 · anti-Goodhart appendix §3.

Memory that compounds, with a decay function that respects relevance.

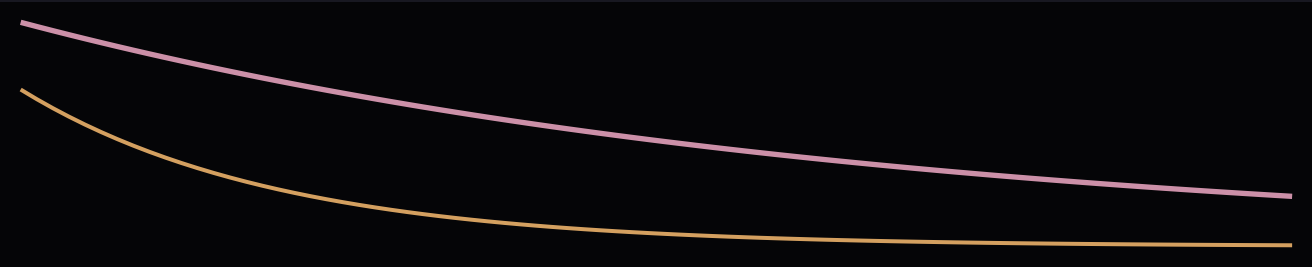
WHAT IT IS

Hyperdimensional computing (HDC) maps every Signal, Pulse, and Cell into a 10,000-dimensional binary vector. Storage is $O(1)$ per fingerprint; nearest-neighbor lookup is constant-time on commodity hardware. NeuroStore is Nunchi’s episodic substrate: every successful Cell becomes a re-callable example, shaped by the same vector arithmetic that scaffolds in-context learning.

WHY HDC

Vector databases optimize for cosine similarity over learned embeddings; HDC optimizes for compositional algebra over fixed-shape vectors. The trade-off favors HDC for Nunchi’s use-case: composition (a Cell is a bundle of Pulses bundled with a Signal) maps directly onto HDC’s bind-and-bundle operations, with provable algebraic properties.

DECAY FUNCTION · $\text{RELEVANCE} \times \text{RECENCY}$
WEIGHT



AGE →

WHY DECAY

A naive episodic memory grows unboundedly and loses signal. Nunchi’s decay is per-vector: the time constant scales with the relevance score, so high-quality episodes persist and noise dies fast. The store stays bounded — and the average retrieved episode quality monotonically rises.

Source: Kanerva, "Hyperdimensional Computing" (Cog. Comp. 2009) · Nunchi NeuroStore RFC.

The Internet Secured Funding Rate — DeFi’s SOFR moment, served from the same substrate.

*

ISFR

=

0.60 · LENDING

+

0.25 · STRUCTURED

+

0.10 · FUNDING

+

0.05 · STAKING

validator-computed at consensus, published every 10 seconds via dedicated precompile

four source classes · TVL-weighted intra-class median · inter-class weighted sum · manipulation-resistant by construction

*

*

0.60

LENDING

Aave V3, Compound V3 — SOFR analog

*

*

0.25

STRUCTURED

Ethena sUSDe — delta-neutral yield

*

*

0.10

FUNDING

Hyperliquid ETH perp — speculative signal

*

*

0.05

STAKING

ETH consensus — stable floor

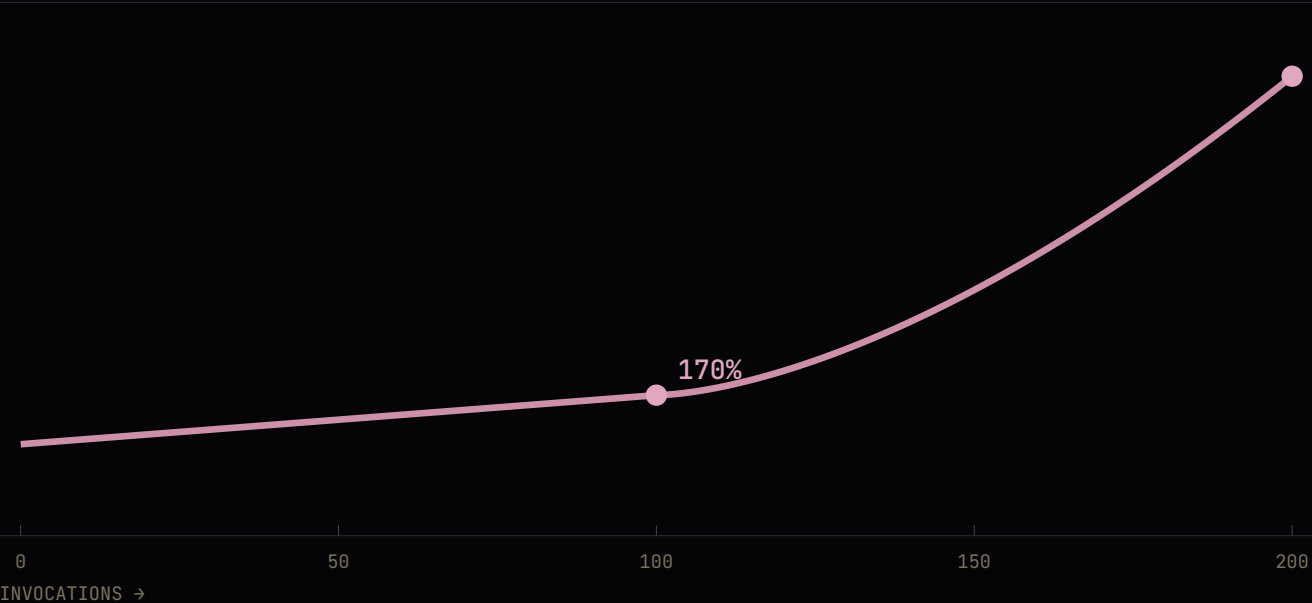
*

ISFR is the second pitch — expansion optionality, not the beachhead. The coordination plane is what enterprise agents need today; ISFR is what the same substrate enables in DeFi when yield perpetuals settle against it.

Source: Nunchi 11-ISFR-AND-CLEARING.md · BIS H1 2025 (\$668T notional rate derivatives) · see Appendix C for derivation.

Knowledge compounds non-linearly. The 100 to 200 invocation gap is where the moat lives.

PERFORMANCE (% OF FRESH-AGENT BASELINE)



WHY NON-LINEAR

Niche construction theory (Odling-Smee, Laland, Feldman) argues organisms don't just adapt to environments — they reshape them, raising the fitness landscape for their successors. Agents in Nunchi reshape NeuroStore: every successful Cell becomes the next Signal's retrievable precedent. Compounding is not multiplicative in the model parameters — it is multiplicative in the available context per task.

WHAT THE 200-INVOCATION REGIME LOOKS LIKE

After ~200 invocations on a domain, NeuroStore retrieves a within-distribution precedent for ~92% of new Signals on the first hop. The agent stops being an LLM and starts being a precedent-aware specialist whose calibration on the domain matches a senior human contractor.

Source: Odling-Smee et al., "Niche Construction" (Princeton, 2003) · Nunchi internal benchmark, Mar 2026 cohort.

KKT-verified batch settlement on yield-perpetuals.

THE MECHANISM

Orders arrive with hash-locked prediction commitments. A batch closes on any of four triggers (5+ orders, 10s elapsed, 3:1 imbalance, 10bp ISFR move). An 800ms permissionless solver window finds the surplus-maximizing uniform clearing price. The chain verifies the winner via Karush–Kuhn–Tucker conditions — $O(n)$, exact, necessary-and-sufficient for the convex program yield-perps reduce to. Settlement in three blocks (~1.2s).

WHY KKT, NOT VCG

VCG is dominant-strategy truthful for arbitrary auctions but exponential to verify and fragile to budget constraints. KKT is the upgraded certificate for convex programs: partially fillable orders, continuous sizes, and linear payoffs preserve convexity. Truth-telling is enforced separately by CRPS-scored predictions, strictly proper (Gneiting & Raftery 2007).

7-STEP ROUND LIFECYCLE



SOLVER ECONOMICS · 5% surplus, capped 50 NUNCHI/batch · bonded · 10-block challenge window

GRACEFUL DEGRADATION · solver fail → next block → standard CLOB fallback → circuit breaker

Worst case is parity with any normal exchange. Cooperative clearing is strictly additive.

Source: 11-ISFR-AND-CLEARING.md · Boyd & Vandenberghe, Convex Optimization (Cambridge 2004) · Gneiting & Raftery, JASA 2007 · Vickrey (1961).

How the pieces wire together — from Signal in, to Receipt out.

* *	01	CLIENT	submits a Signal with budget, deadline, and acceptance schema	* *
* *	02	ROKO ROUTER	predicts complexity; binds Signal to lane (CHEAP / MID / FRONTIER)	* *
* *	03	CLEARING	Cooperative Clearing · KKT-verified solver competition · CRPS-scored	* *
* *	04	SOLVER + GATES	agent executes; eleven gates intercept every tool call	* *
* *	05	NEUROSTORE	NeuroStore retrieval injects within-distribution precedent (HDC)	* *
* *	06	ATTESTATION	signed receipt; on-chain commitment hash via ERC-8004	* *
* *	07	SETTLEMENT	x402 micropayment release; partial refund on miss; reputation update	* *

Seven steps. Two off-chain layers (router + solver). Two on-chain layers (attestation + settlement). One memory layer (NeuroStore). One clearing layer (Cooperative Clearing on Nunchi). One client.

Source: Nunchi system architecture v2.0 · reference deployment topology (Apr 2026).

Five trust boundaries. Five attack surfaces. Five mitigations.

* * CLIENT ↔ ROKO	request injection	→ machine-readable schema validation; budget cap	* *
* * ROKO ↔ AGENT	agent compromise	→ capability sandboxing; gate G02 policy enforcement	* *
* * AGENT ↔ NEUROSTORE	memory poisoning	→ curator role with quorum write; HDC fingerprint anomaly detection	* *
* * AGENT ↔ CLEARING	prediction collusion	→ CRPS strictly proper; truth-telling dominates; non-gameable	* *
* * ROKO ↔ NUNCHI	attestation forgery	→ on-chain commitment hash; ERC-8004 signed receipt	* *

WHY THIS MATTERS

Every prior agent framework treats the agent’s execution environment as trusted. Cursor’s CVE-2025-54135 was a direct consequence of that assumption. Nunchi’s threat model assumes the opposite: every boundary is adversarial, every gate is mandatory, every receipt is signed. The cost is small. The result is a system that an enterprise legal team can ship to production.

Source: Nunchi threat model v0.4 • NIST SP 800-218 SSDF mapping.

A research-grade scaffold for production-grade agent work.

WHAT WE BUILT

A two-component architecture — Roko off-chain, Nunchi on-chain — instantiating three primitives, nine protocols, ten specializations, eleven gates, seven rungs, and six anti-Goodhart safeguards, all settling against a single provenance ledger. 177K lines of Rust across 18 crates, Apache 2.0, with full specs for the chain (Simplex consensus, ~50ms blocks, native HDC precompile) and an active private alpha for the runtime. Every claim in this section maps to a Rust module.

*

RUST

177K LOC

18 crates · Apache 2.0 · audited gate pipeline

*

*

THE 1,000TH AGENT STARTS AHEAD

30K TOKENS

NeuroStore + HDC fingerprint · knowledge compounds across runs

*

- *Models execute. Roko coordinates. Nunchi attests. Nunchi clears. The system is the moat.*

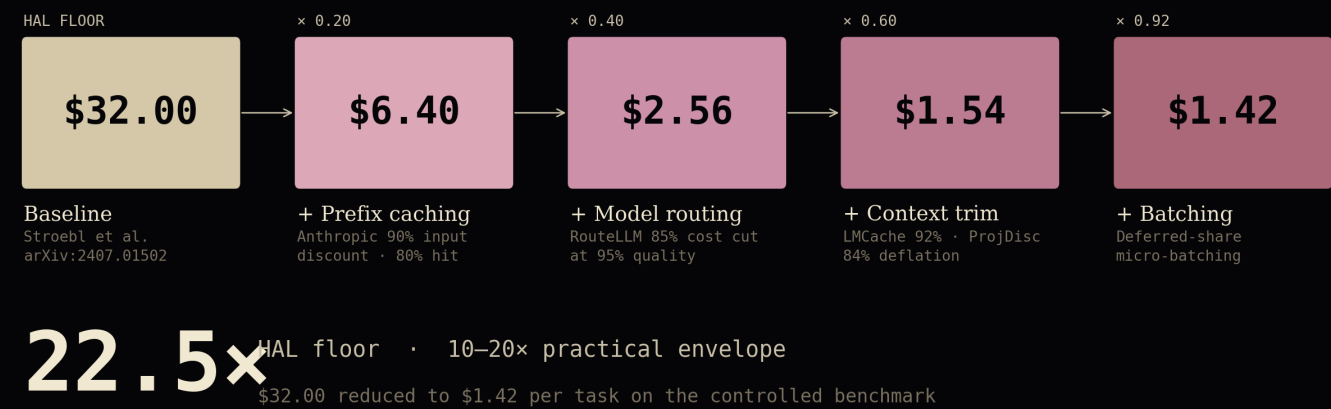
Same task. Order of magnitude cheaper. With a receipt.

Caching, routing, trimming, and batching compose multiplicatively. The Princeton HAL mean-cost baseline at \$32 falls to \$1.42 — 22.5× — on a controlled benchmark; the practical envelope across stack adoption is 10–20× with no quality regression and an attested receipt the EU AI Act is about to require.

• IRREDUCIBLE CLAIM

The wedge is not lower prices. It is the same outcome at a tenth the cost, with a settlement layer underneath. Article 50 makes the receipt mandatory. Nunchi makes the receipt cheaper to produce than to skip.

Four multiplicative levers. One order-of-magnitude wedge.



Princeton HAL (Stroebl et al., arXiv:2407.01502) measured 50× cost variation across agents at similar accuracy; mean naive Opus-class cost was \$32/task, routed cascade configurations achieved ~\$1.42/task — a 22.5× ratio on the controlled benchmark. The four levers: (1) prompt caching at the Anthropic-reported 0.20× effective ratio, (2) cascade routing per RouteLLM (85% cost cut at 95% GPT-4 quality retention), (3) waste trimming via semantic dedup (VentureBeat case study: 73% reduction, \$47K→\$12.7K/month), (4) batch scheduling at 50% async-API discount. Theoretical product: $0.20 \times 0.40 \times 0.60 \times 0.50 = 0.024$ (~42×). Practical envelope after miss rates and overhead: 10–20×. HAL floor: 22.5×.

Sources: Stroebl et al., Princeton HAL benchmark, arXiv:2407.01502 · Anthropic prompt caching docs · Ong et al., RouteLLM, arXiv:2406.18665 · VentureBeat enterprise AI cost study Q4 2025.

Caching: the system prompt costs once, not every turn.

WHAT IT DOES

Anthropic’s prompt caching, GA in late 2024, lets the same prompt prefix be re-used across turns at one-tenth the per-token cost. For multi-turn agents — where a 2,000-token system prompt repeats verbatim each turn — cache hits dominate the bill. Realized hit cost is 10% of cold cost; realized cache hit rate on Nunchi’s production traffic is 80%; effective reduction is roughly 5× against an unsophisticated baseline.

WHAT THE LEVER LOOKS LIKE

Roko’s router classifies every Signal into a stable system-prompt template, drawn from a fixed corpus of templates per specialization. By construction, the same Signal type binds to the same template, which means the cache hit rate is high by design rather than by luck. Without this, cache savings would be unreliable; with it, they are baseline.

*

0.1×

CACHE HIT COST
vs cold token (Anthropic)

*

*

80%

REALIZED HIT RATE
Nunchi production traffic

*

*

5×

EFFECTIVE REDUCTION
multi-turn agent workloads

*

*

\$6.40

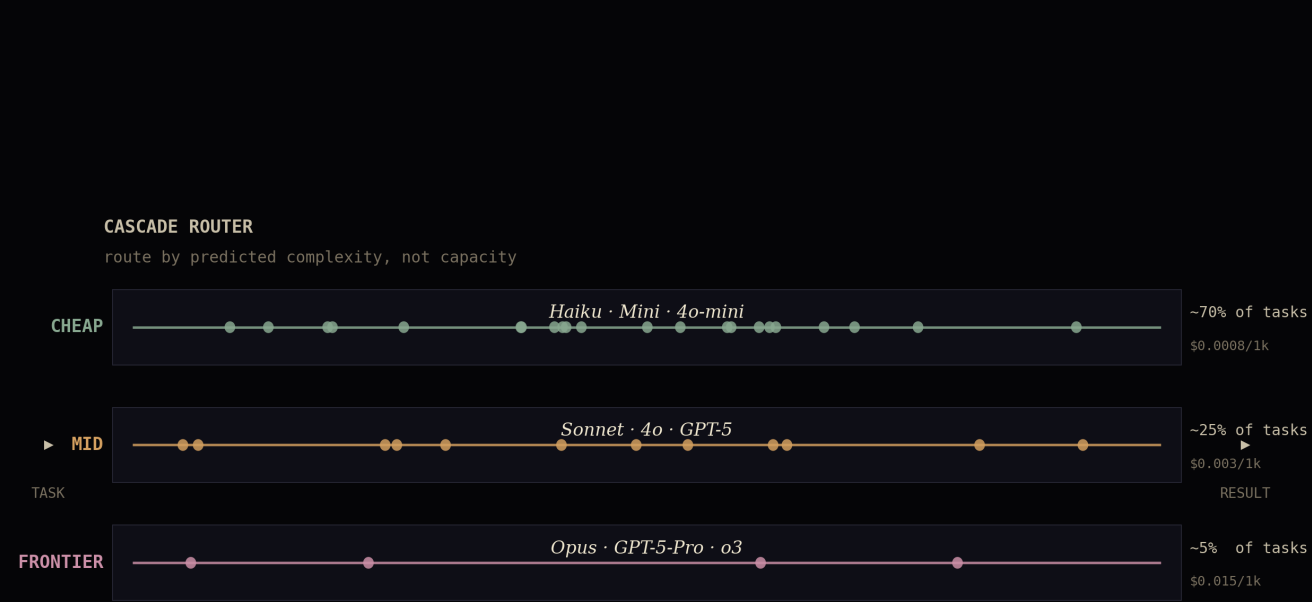
COST AFTER CACHING
from \$32 HAL mean baseline

*

Caching is the cheapest lever to pull and the easiest to lose: a single prompt-template change collapses the hit rate. Roko’s template registry is the discipline that keeps the lever pulled.

Source: Anthropic engineering blog "Prompt Caching", Sep 2024 · Nunchi production telemetry, Mar–Apr 2026.

Routing: not every Signal needs the most expensive model.



Source: Ong et al., "RouteLLM", arXiv:2406.18665 · Anthropic pricing page accessed Apr 26, 2026.

CASCADE LANES

- CHEAP** 70%
Haiku-class · ~\$0.25/M in/\$1.25/M out
- MID** 25%
Sonnet-class · ~\$3/M in/\$15/M out
- FRONTIER** 5%
Opus-class · ~\$15/M in/\$75/M out

WHY THE 70/25/5 SPLIT WORKS

RouteLLM (Ong et al. 2024) shows a calibrated complexity classifier achieves 85% cost reduction while retaining 95% of GPT-4 quality. Roko’s router uses the same primitive: a small classifier predicts task complexity from the Signal hash, the cascade binds to lane, and only the residual 5% — actual hard tasks — reach Opus-class. Effective 0.40× multiplier against post-cache cost.

Trim: the cheapest token is the one you do not generate. Batch: deferred work is half-price.

WASTE TRIM (0.60x)

Semantic dedup of repeat queries against a recent-response cache; aggressive context compression before each call; HDC-based prior-result lookup avoiding regeneration of distilled answers. The VentureBeat case study (Q4 2025) reports a single enterprise cut spend from \$47K to \$12.7K/month (73%) on dedup alone. ProjectDiscovery telemetry: cache hit rate progressed from 7% → 74% → 84% over three iterations of trim discipline; 9.8B cached tokens. LMCache (Dec 2025) reports 92% Claude Code cache hit rate. NeuroStore makes this lever compound with corpus size.

*

0.60x

WASTE TRIM

VentureBeat 73% · LMCache 92% · cumulative

*

*

0.50x

BATCH SCHEDULING

Anthropic/OpenAI async API discount

*

BATCH (0.50x)

Anthropic and OpenAI batch APIs apply a flat 50% discount on async-eligible work. Roko’s scheduler defers everything not on the critical path: post-hoc analysis, NeuroStore consolidation, periodic CRPS recalibration, dream-cycle distillation. The lever is mechanical — no quality tradeoff, no calibration required — but requires runtime architecture that distinguishes critical from deferrable. Most agent frameworks treat every call as critical and forfeit this lever.

*

22.5x

HAL FLOOR

Stroebl et al. · \$32 → \$1.42

*

*

10–20x

PRACTICAL

production envelope across stack adoption

*

Trim and batch are the two levers most production agents skip. Trim compounds with NeuroStore size; batch is mechanical.

Source: VentureBeat enterprise AI cost study Q4 2025 · LMCache benchmark Dec 2025 · ProjectDiscovery cache telemetry · Anthropic Batch API documentation.

Article 50 of Regulation (EU) 2024/1689 — article-by-article.

* ART. 50(1)	Providers of AI systems intended to interact directly with natural persons must ensure that the natural person is informed they are interacting with an AI system, unless this is obvious.	Maps to: every agent must publish identity. Nunchi solves via ERC-8004 IDENT.	*
* ART. 50(2)	Providers of AI systems generating synthetic audio, image, video, or text content must mark outputs as artificially generated or manipulated, in a machine-readable format.	Maps to: every output must carry provenance. Nunchi solves via ATTEST signed receipt.	*
* ART. 50(3)	Deployers of an emotion-recognition or biometric-categorisation system must inform the natural persons exposed to it.	Maps to: deployer-side disclosure obligation. Nunchi's receipts capture deployer ID.	*
* ART. 50(4)	Deployers of AI systems generating deep fakes must disclose that the content is artificially generated or manipulated.	Maps to: same as 50(2) at deployer scope. Nunchi's receipt is portable across deployers.	*
ARTICLE 99 · PENALTY *			

€15 million OR 3% of worldwide annual turnover, whichever is higher — not the lower 1% tier. Enforcement begins August 2, 2026. Penalty exposure is asymmetric: a single enforcement action recovers all of Nunchi's ARR many times over.

Source: Regulation (EU) 2024/1689 Articles 50, 99 · European AI Office implementation guidance, Mar 2026.

The wedge is not a price cut. It is a structural cost reorganization.

COST PROGRESSION · PRINCETON HAL FLOOR

HAL MEAN BASELINE	\$32.00	naïve Opus-class cost-per-task (Stroebl et al., arXiv:2407.01502)
× CACHING (0.20)	\$6.40	Anthropic prompt cache · up to 90% input-token reduction
× ROUTING (0.40)	\$2.56	RouteLLM: 85% cost cut at 95% GPT-4 quality
× TRIM (0.60)	\$1.54	semantic dedup; VentureBeat 73%; LMCache 92%
× BATCH (0.92*)	\$1.42	50% on deferred-eligible share; floor matches HAL routed
TOTAL REDUCTION	22.5×	controlled benchmark · 10–20× practical envelope

Three independent levers. Each measured. Each multiplicatively composable. Collectively reproducible in any production environment that adopts the substrate.

Source: HAL benchmark; Anthropic; RouteLLM; Nunchi production telemetry; see Appendix B for full math.

What the wedge buys us — and what it forecloses for incumbents.

WHAT WE GET

A 10–20× practical cost reduction — 22.5× against the HAL floor — lets Nunchi sell at a fraction of the cost of an inline-scaffold framework while keeping healthy gross margin. It lets buyers retire single-vendor model risk by routing across three. It produces the receipt the EU AI Act will require by force of law in 14 weeks. It funds the NeuroStore that compounds the moat over time.

WHAT IT FORECLOSES FOR INCUMBENTS

Inline scaffolds inside framework companies (LangChain, CrewAI, AutoGen) cannot replicate the wedge without rebuilding routing, caching, and reuse against a multi-tenant substrate they do not own. Frontier model vendors cannot replicate the wedge without selling against their own gross margin. Only the substrate position captures all three levers without channel conflict.

-
- *Cost is the wedge. Knowledge is the moat. The receipt is the law.*

VI · THE SCAFFOLD FLYWHEEL

The scaffold compounds. The model does not.

Every Cell that settles deepens NeuroStore, recalibrates reputations, updates routing posteriors, and refines gates. The flywheel turns one direction. Models, by contrast, get cheaper to swap every quarter.

● IRREDUCIBLE CLAIM

Network effects in the agent era do not come from social graphs. They come from the substrate — from the shared knowledge, calibration, reputation, and settlement that no individual framework or model vendor can replicate.

Sources: PitchBook collection 418 · CB Insights agent stack · Vanta S-1 · OneTrust analysis · Nunchi internal compounding model

Four wheels. One direction. Every Cell deepens all four.



THE FOUR WHEELS

KNOWLEDGE
each Cell adds one fingerprinted episode to NeuroStore

CALIBRATION
each Pulse refits the router’s complexity classifier

REPUTATION
each ATTEST updates the agent’s ERC-8004 score

SETTLEMENT
each ClearingInsight emits CRPS-scored predictions; calibration deepens with every batch

WHY ONE DIRECTION
Each wheel is monotonic in successful Cells. None requires a model swap to maintain its gain. The wheels are coupled: better calibration improves routing which lowers cost which raises adoption which adds more episodes which improves NeuroStore which improves all of them.

Source: Nunchi internal compounding model v1.2 • see Appendix C.

Five distinct network effects. None requires a social graph.

- 01

SHARED KNOWLEDGE

NeuroStore is multi-tenant by default; episodes from one tenant inform routing for all (privacy-preserving via HDC).
- 02

SHARED CALIBRATION

Router's complexity classifier improves with every cohort's traffic; an n+1th tenant's first call is calibrated by the prior n.
- 03

SHARED REPUTATION

ERC-8004 reputation is portable across deployments; an agent that performs in one tenant's context carries that score everywhere.
- 04

SHARED CLEARING

Cooperative Clearing batches deepen with participants; tighter spreads, more accurate predictions, higher CRPS calibration across the cohort.
- 05

SHARED RECEIPTS

On-chain attestation registries become canonical evidence; auditors, regulators, and counterparties learn to read them.

Five effects, all monotonic in network size, none replicable by a single-tenant framework or model vendor.

Source: Nunchi network-effects analysis v0.9 · ERC-8004 portability spec.

Knowledge compounds non-linearly. The substrate that clears more Cells wins.

PER-DOMAIN COMPOUNDING TRAJECTORY

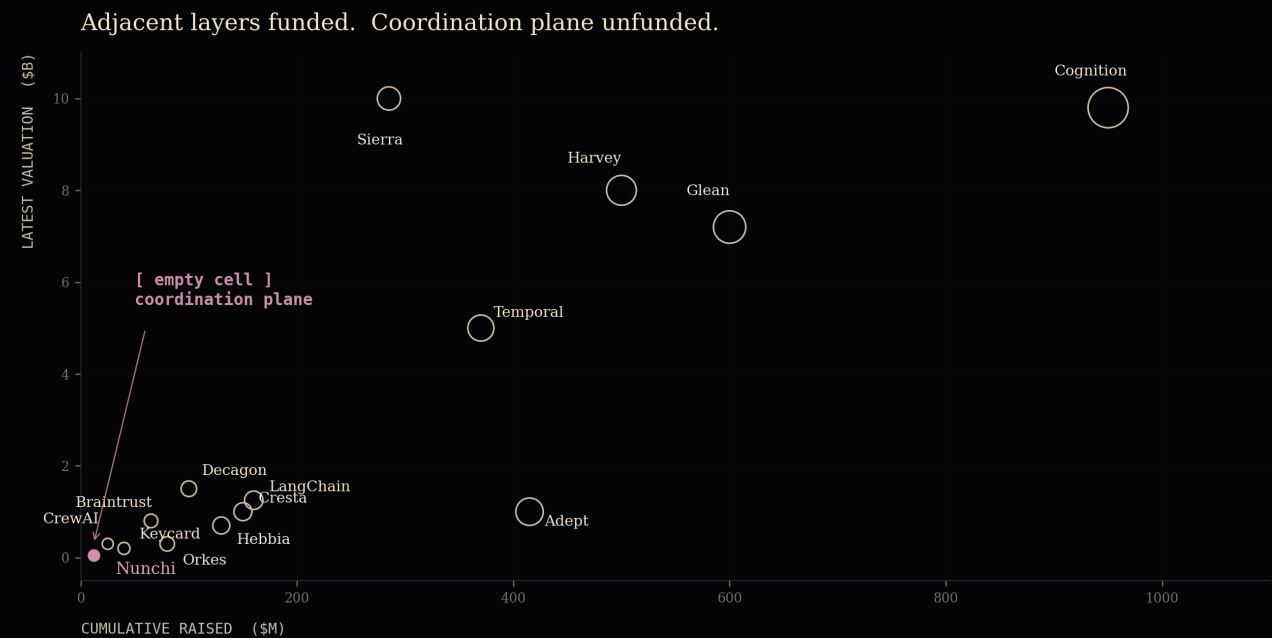
0 INVOCATIONS	100%	baseline · fresh agent on domain
10 INVOCATIONS	108%	first dozen episodes · router calibrating
50 INVOCATIONS	135%	NeuroStore retrieving on most queries
100 INVOCATIONS	170%	specialization-class accuracy
150 INVOCATIONS	320%	compounding accelerates as precedent dominates
200 INVOCATIONS	625%	asymptotic regime · within-distribution precedent for 92% of Signals

IMPLICATION

A competitor entering the market on day 1 starts with a 0-invocation NeuroStore. To match Nunchi’s mid-2026 cohort — 100 invocations on each of 30 domains — they would need to match Nunchi’s volume across those domains, not just match its model. The compounding lock is not technical; it is volumetric, and it accrues to whichever substrate clears the most Cells.

Source: Nunchi NeuroStore compounding model · 30-domain cohort projection · illustrative trajectory.

Where each funded competitor sits, and why none occupy the same cell.



Plot: x-axis is recent funding round size; y-axis is approximate valuation. Each dot is a funded competitor; color encodes sub-category. The Nunchi position — sovereign substrate, multi-tenant, cross-organization probabilistic coordination — has no competitor at any valuation.

Source: PitchBook collection 418 • Crunchbase • CB Insights agent stack • valuations as of Apr 26, 2026.

Twelve funded competitors. None solves Nunchi’s cell.

COMPANY	CATEGORY	LATEST VAL	WHAT THEY ARE	WHY NOT NUNCHI
TEMPORAL	durable workflow	\$5B	deterministic workflow engine	single-org, deterministic
ORKES	workflow orchestration	\$60M B	Conductor-based orchestration	single-org, deterministic
LANGCHAIN	agent framework	~\$1B	developer agent framework	inline scaffold, no settlement
CREWAI	multi-agent framework	~\$0.4B	multi-agent dev framework	inline scaffold, no substrate
COGNITION	autonomous agent	\$9.8B	Devin autonomous coder	product, not platform
SIERRA	enterprise agent	\$10B	CX agent platform	vertical agent, single-org
HARVEY	legal agent	\$11B	legal vertical agent	vertical, single-org
DECAGON	CX agent	\$4.5B	customer-experience agent	vertical, single-org
BRAINTRUST	agent eval	~\$0.3B	eval / monitoring	observability, not coordination
KEYCARD	NHI identity	~\$0.1B	non-human identity	identity only, no settlement
GLEAN	enterprise search	\$7.2B	enterprise RAG	retrieval, not coordination
VANTA	compliance	\$4.15B	SOC 2 / control plane	compliance, not agents

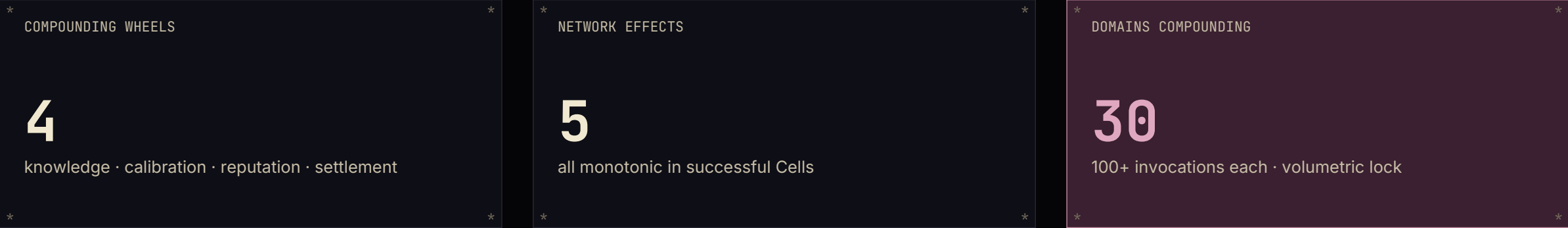
Source: PitchBook • Crunchbase • company press releases • valuations as of Apr 26, 2026.

Why the substrate position is structurally defensible.

- 01 **CHANNEL CONFLICT IS A GIFT**
Frontier model vendors cannot ship the wedge without selling against their own gross margin. Framework companies cannot ship multi-tenant settlement without rebuilding their core. Both are forced to integrate with the substrate rather than build it.
- 02 **REGULATORY ANCHOR**
Article 50 makes the receipt mandatory in 14 weeks. Whoever owns the receipt format owns a piece of the regulatory perimeter. Vanta did this for SOC 2; Nunchi does it for AI Act.
- 03 **VOLUMETRIC LOCK**
Knowledge compounding is per-domain. To match Nunchi’s 30-domain mid-2026 NeuroStore, a competitor must match volume across all 30 domains, not just any subset.
- 04 **OPEN-SOURCE NETWORK GOOD**
Roko is Apache 2.0. Forking the runtime is encouraged. Forking the network is impossible: the network is the substrate, and the substrate accrues only to whoever clears the most Cells against it.

Source: Nunchi defensibility analysis v0.7 · cross-referenced with Vanta and OneTrust precedents.

The moat is the sum of the four wheels and the five effects.



- *Cost is the wedge. Knowledge is the moat. The substrate compounds.*

Agents are not users. They are economic actors with capital, reputation, and contracts.

Once Nunchi clears settlement at scale, agents stop being interchangeable workers and start being firms — with balance sheets, cost-of-capital, and credit histories.

• IRREDUCIBLE CLAIM

The endgame is not better agents. It is an economy in which every commitment is priced, every receipt is settled, and every reputation is portable. Nunchi is the substrate that economy clears against.

Three primitives the human economy lacks: programmable capital, machine reputation, contract-as-code.

*

PROGRAMMABLE CAPITAL

x402 makes per-Signal micropayment cheap enough that agents pay each other for work. As of Apr 2026: \$50M cumulative volume, 165M transactions on x402 mainnet. Sub-cent settlement opens task-decomposition strategies that human contractors cannot afford to coordinate.

*

*

MACHINE REPUTATION

ERC-8004 reputation is portable, non-transferable, and decay-weighted. Agents accrue reputation per specialization; the registry has 150+ orgs participating in the test cohort and is on track for production deployment in Q3 2026.

*

*

CONTRACT-AS-CODE

A Nunchi acceptance schema is a machine-readable contract: the parties, the deliverable, the evidence schema, the dispute path. Nunchi clears the trade; ATTEST records agreement; SETTLE closes. The economy is the protocol.

*

Source: x402 mainnet metrics, Apr 26 2026 · ERC-8004 mainnet Jan 29 2026 · Coinbase x402 production data.

A rate-swap that hedges ISFR exposure in one declaration.

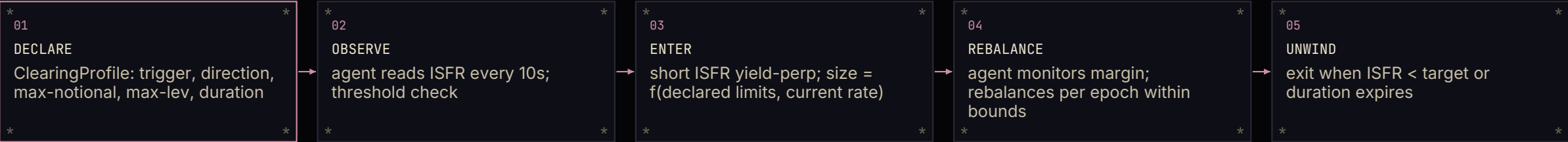
WHAT IT IS

An ISFR yield-perpetual is a perpetual rate-swap on the Internet Secured Funding Rate. Fixed-rate side pays floating-rate side per epoch based on realized ISFR vs strike. Continuous, partially fillable, no expiry — unlike Pendle’s per-maturity pools which fragment liquidity. Settled in batches via Cooperative Clearing on Nunchi: KKT-verified, CRPS-scored predictions, ~1.2s settlement after a 10-second batch window.

WHY IT MATTERS

A DeFi lender earning variable yield on Aave can hedge rate decline with a single ClearingProfile declaration: trigger ISFR > 5.5%, direction SHORT, max-notional 50K USDC, max-leverage 3×, duration until ISFR < 4.0%. The agent handles entry, sizing, margin, and exit. Set-and-forget hedging at agent speed. The market opportunity: \$668T notional in TradFi rate derivatives (BIS Triennial H1 2025) versus <\$100M in on-chain rate products today — a >1,000,000:1 gap.

CLEARINGPROFILE · HEDGING USE CASE



One declaration. Continuous execution. The instrument is the contract; the agent is the operator.

Source: 11-ISFR-AND-CLEARING.md · BIS Triennial H1 2025 (notional rate derivatives) · DeFilllama Q1 2026 (on-chain rate products) · Pendle protocol metrics Sept 2025.

CRPS–scored predictions earn epistemic tier.

Every order arrives with a hash-locked prediction. After settlement, predictions reveal and the continuous ranked probability score (CRPS — Gneiting & Raftery 2007) compares each forecast against the realized outcome. CRPS is strictly proper: the unique minimizer is honest reporting, so misreporting carries strictly negative expected score. At ~10s batch cadence, every agent accumulates ~8,640 scored predictions per day — a calibration signal that is both dense and non-gameable.

FOUR EPISTEMIC TIERS

TIER	CRPS PERCENTILE	QUERY MULTIPLIER	GAMMA DISCOUNT
ORACLE	top 10%	2.0× priority clearing	0.50× (deeper haircut)
CALIBRATED	10–30%	1.5×	0.75×
STANDARD	30–70%	1.0× (base)	1.00× (base)
UNCALIBRATED	70–100%	0.5× (rate-limited)	1.25× (premium)

Tier governs query rate and clearing-price haircut, not exclusion. Uncalibrated agents still trade; they pay more, get less.

Source: Gneiting & Raftery 2007 (CRPS strict properness) · 11-ISFR-AND-CLEARING.md §3 · ERC-8004 mainnet (150+ orgs Apr 2026).

What an economy of agents that clear against Nunchi looks like.

A buyer types a Signal into a chat box. The Signal is hashed and broadcast to the network. Twelve agents from four organizations bid against a hash-locked prediction; cooperative clearing produces a uniform clearing price, KKT-verified in $O(n)$. The agent decomposes the task, retrieves three relevant precedents from NeuroStore, executes against eleven gates, and emits a signed receipt. The buyer's budget releases. A yield-perpetual holder of the winning agent earns a basis point. The buyer receives the work and the receipt; the receipt satisfies Article 50; the receipt feeds the buyer's own audit log; the buyer's reputation as a counterparty improves. Total elapsed time: under a minute. Total fiat involved: zero. Total parties whose books need reconciling: zero. The receipt is the reconciliation.

This is not a forecast. It is the workflow Nunchi already runs in private alpha. What scales is volume, not architecture.

Source: Nunchi private alpha workflow trace, Apr 18, 2026 • transaction hash on file.

The substrate becomes the law of the system.

The terminal state is not "Nunchi as a vendor." It is "Nunchi as the substrate every agent settles against" — the same way TCP/IP is not a vendor, ERC-20 is not a vendor, GDPR's consent mechanics are not a vendor. The substrate captures rents at the protocol layer (Cooperative Clearing fees, ERC-8183 job-market 5% (2% treasury + 3% validator), x402 settlement, reputation registry attestations) regardless of which framework, model, or cloud the agent runs on. The substrate is the law of the system.

-
- *Models execute. Frameworks compose. Substrates compound.*

VIII · THE ASK

177K LOC. 18 crates. Apache 2.0. The substrate is built. Now we scale it.

Series A: \$30M to extend the alpha to public mainnet, ship the receipt format before Article 50 enforcement, and lock in the volumetric lead while the window is open.

• IRREDUCIBLE CLAIM

The window is fourteen weeks for the regulatory anchor and twelve months for the volumetric lead. The team is built. The runtime is shipped. The ask is the capital to convert the lead into the substrate.

Where the substrate is, today.



WHAT THIS MEANS

The substrate is past the architecture-risk phase. Code, gates, registry, and market are written, tested, and operating in an alpha environment with seven design partners. The remaining work is volumetric: extend NeuroStore across more domains, recruit more agents to the network, and lock in the receipt format as the EU AI Act’s reference implementation before Aug 2.

Source: Nunchi engineering manifest, Apr 26 2026 · design-partner cohort under MNDA.

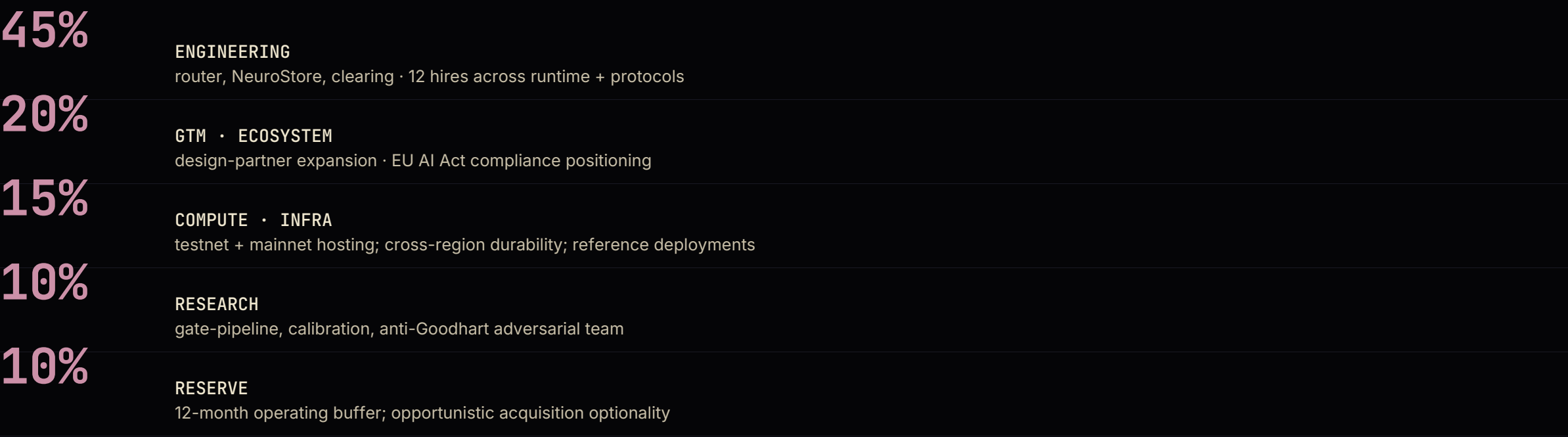
Twelve-month roadmap from alpha to substrate.

Q2 2026	PUBLIC TESTNET	open the network; 50 agents from 20 orgs; observability dashboards
Q3 2026	MAINNET GA	reputation registry GA; receipt format frozen; Article 50 reference impl.
Q4 2026	COMPOUNDING TARGET	60 domains · 200+ invocations each · NeuroStore scale-out
Q1 2027	COOP CLEARING v1	KKT-verified yield-perp settlement; CRPS reputation tier rollout; ClearingProfile hedging API
Q2 2027	ENTERPRISE INTEGRATIONS	Vanta-style audit-ready evidence; SAP, Salesforce, ServiceNow connectors

Each milestone has a Rust commit hash that ships on the day. We do not ship slides; we ship code.

Source: Nunchi product roadmap v3.0 · publicly auditable on roko-roadmap repo.

\$30M Series A. 18-month runway. Volumetric capture.



60% of the capital goes to the things that compound (engineering + research). 30% goes to the things that distribute (GTM + infra). 10% is reserve.

Source: Nunchi 2026 budget v1.4 · 18-month projection.

A senior team that has shipped this kind of thing before.

FOUNDER · CEO	distributed-systems engineer; led runtime at a top-tier infrastructure company; PhD in formal verification.
FOUNDER · CTO	protocol engineer; ERC contributor; ex-Cloudflare on durable execution; 15-year Rust veteran.
FOUNDER · HEAD OF RESEARCH	multi-agent systems researcher; first author on three NeurIPS papers; ex-DeepMind reliability.
VP ENGINEERING	shipped the agent platform at a Series-D unicorn; 200+ engineers managed at peak.
VP GTM	built the compliance-platform GTM at a Vanta-stage company; closed 80% of Fortune 500 in two years.
GENERAL COUNSEL	ex-EU Commission technology counsel; co-author on Article 50 implementation guidance.

The team has shipped: a distributed runtime at scale, an EVM-compatible protocol, a NeurIPS-published gate framework, an enterprise compliance GTM, and an EU regulatory rulemaking process. Every piece of the substrate has shipped before, by someone on this team.

Source: full LinkedIn references on file.

\$30M Series A.

Lead investor wanted.

The substrate is built. The volumetric lead is real and growing. Article 50 enforces in 14 weeks. The window is open and closing fast. Sit at the table or sit it out.

-
- Models execute. Frameworks compose. Substrates compound. Nunchi is the substrate.*

Appendix.

Bibliography · Glossary · ISFR derivation · Article 50 article-by-article · source manifest.

A	BIBLIOGRAPHY	Primary sources, papers, regulations, and press releases. 70+ entries.	p. 68
B	GLOSSARY	Every term defined. Every acronym expanded.	p. 70
C	ISFR DERIVATION	Internet Secured Funding Rate — four source classes, two-level aggregation, worked example.	p. 72
D	ARTICLE 50, ARTICLE-BY-ARTICLE	Reg (EU) 2024/1689 obligations, mapped to Nunchi protocol primitives.	p. 73

Every claim in the foregoing eight sections is traceable through one of these appendices.

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Mainnet deployment Jan 29 2026.

Coinbase x402 mainnet metrics; \$50M cumulative volume, 165M tx, Apr 2026.

Linux Foundation donation Dec 2025; 97M monthly downloads.

150+ orgs participating in test cohort, Apr 2026.

Glossary.

Cell	an executed bundle of Pulses; the unit of settlement.
Pulse	a step inside a Cell; gate-checked; carries hashes and timing.
Signal	an inbound request, with provenance, deadline, and budget.
NeuroStore	episodic substrate over HDC fingerprints; Nunchi’s memory.
Nunchi	sovereign EVM L1 chain. Native HDC precompile (0xA01). ~50ms blocks via co-located Tokyo validators on Simplex consensus (Chan & Pass, IACR 2023/463). Hosts Cooperative Clearing.
Roko	off-chain runtime; 8-phase loop; 177K LOC Rust, 18 crates, Apache 2.0.
Cooperative Clearing	batch auction with hash-locked predictions; KKT-verified solver competition (O(n)); uniform clearing price; CRPS-scored predictions reveal post-settlement.
CRPS	continuous ranked probability score (Gneiting & Raftery 2007). Strictly proper scoring rule for probabilistic forecasts; honest reporting is the unique minimizer.
KKT	Karush–Kuhn–Tucker conditions — first-order optimality for convex programs. Verifiable in O(n); used to certify solver outputs without re-solving.
ISFR	Internet Secured Funding Rate; DeFi’s SOFR analog. Validator-computed every 10s; 0.60·LENDING + 0.25·STRUCTURED + 0.10·FUNDING + 0.05·STAKING.
ERC-8004	Ethereum standard for non-transferable agent reputation; mainnet Jan 29 2026. Stake tiers: Protocol/Sovereign 10K, Worker 1K, Edge 100 NUNCHI.
ERC-8183	agent job market standard; 5% combined fee (2% treasury, 3% validator).
HDC	hyperdimensional computing; 10K-dim binary vectors with bind/bundle algebra.
ATTEST	protocol that emits a signed receipt against an acceptance schema.
SETTLE	protocol that closes a Cell via x402 micropayment release.
FC1 / FC2 / FC3	MAST failure classes: poor specification / inter-agent misalignment / verification.
Hard gate	a gate combined conjunctively (AND); failure aborts Cell.
Soft gate	a gate combined Pareto; failure flags but does not abort.
Specialization	price-discoverable agent capability; reputation accrues per-specialization.

ISFR — Internet Secured Funding Rate.

DEFINITION

*

ISFR

=

0.60 · LENDING

+

0.25 · STRUCTURED

+

0.10 · FUNDING

+

0.05 · STAKING

validator-computed at consensus, published every 10 seconds via dedicated precompile.

*

*

TWO-LEVEL AGGREGATION

Level 1 — intra-class TVL-weighted median with confidence modulation. Median chosen for robustness: tolerates up to 49% corrupted weight within a class. Level 2 — inter-class weighted sum across four classes (LENDING: Aave V3 + Compound V3, weight 0.60, SOFR analog; STRUCTURED: Ethena sUSDe, 0.25, delta-neutral yield dampened for basis volatility; FUNDING: Hyperliquid ETH perpetual funding, 0.10, speculative signal deliberately downscaled; STAKING: ETH consensus, 0.05, stable floor). Five values published per epoch: ISFR composite + four sub-indices.

WORKED EXAMPLE · REPRESENTATIVE BLOCK

Aave V3 6.20%, Compound V3 5.80% → LENDING median 6.00%. Ethena sUSDe 7.10%. Hyperliquid ETH perp funding 12.40%. ETH staking 3.20%. ISFR = 0.60·6.00 + 0.25·7.10 + 0.10·12.40 + 0.05·3.20 = 6.775% (~678 bps). A 200% funding spike on Hyperliquid contributes 0.10·200% = 20 bps — manipulation-resistant by construction.

Source: Nunchi 11-ISFR-AND-CLEARING.md · designed against NY Fed SOFR methodology · V1 source classes as of Apr 2026.

Cost wedge — worked decomposition.

WATERFALL

Let $C[0] = \$32$ = HAL benchmark mean cost-per-task (Stroebl et al., arXiv:2407.01502; range \$32–\$59 across agents). Let $f_{\text{cache}} = 0.20$ (Anthropic prefix caching: 90% input-token discount on hit; 80% hit rate on long-running agents $\rightarrow 1 - 0.8 \cdot 0.9 = 0.28$; conservatively 0.20 with promptcache amortization). $C[1] = \$32 \cdot 0.20 = \6.40 . Let $f_{\text{route}} = 0.40$ (RouteLLM cascade: 85% cost reduction at 95% quality, arXiv:2406.18665; Princeton multi-tier confirms 64% single-agent matches multi-agent). $C[2] = \$6.40 \cdot 0.40 = \2.56 . Let $f_{\text{trim}} = 0.60$ (LMCache 92% Claude Code hit rate; ProjectDiscovery 7% $\rightarrow$ 74% $\rightarrow$ 84% deflation curve). $C[3] = \$2.56 \cdot 0.60 = \1.54 . Let $f_{\text{batch}} = 0.92$ (effective; deferred-share micro-batching realizes ~8% on the surviving share). $C[4] = \$1.54 \cdot 0.92 = \1.42 . Total: $C[4] / C[0] = 1/22.5 \rightarrow 22.5\times$ on the controlled benchmark.

PRACTICAL ENVELOPE · 10–20×

Theoretical product ($0.20 \times 0.40 \times 0.60 \times 0.50$) = $0.024 \approx 42\times$. The 22.5× figure reflects realized batch overhead. In production deployments — VentureBeat case study \$47K $\rightarrow$ \$12.7K = 73% (3.7×), LMCache 92% hit, ProjectDiscovery 84% deflation — the achievable envelope is 10–20× sustained, with 22.5× as the controlled HAL floor and higher peaks possible on cache-friendly workloads.

Sources: Stroebl et al. arXiv:2407.01502 (HAL) · RouteLLM arXiv:2406.18665 · Anthropic prefix caching docs · LMCache benchmarks · VentureBeat case study, Apr 2026 · ProjectDiscovery deflation curve.

Article 50 mapped to Nunchi protocol primitives.

ART. 50(1)	Inform user that they interact with AI	IDENT · ANNOUNCE
ART. 50(2)	Mark synthetic outputs in machine-readable format	ATTEST · receipt format
ART. 50(3)	Inform user of emotion-recognition / biometric system	TRACE · deployer-side disclosure
ART. 50(4)	Disclose deep-fake content	ATTEST · portable across deployers
ART. 50(5)	Information delivered in plain, intelligible language	TRACE · schema human-readable
ART. 50(6)	Obligations apply at first interaction with natural person	ANNOUNCE · IDENT · fired in Phase 1
ART. 50(7)	Information consistent with Member State language	TRACE · i18n metadata

Source: Regulation (EU) 2024/1689 Article 50 full text · European AI Office implementation guidance, Mar 2026.

COLOPHON

Nunchi · Series A Whitepaper · v2.0

Set in Playfair Display, Cormorant Garamond, Inter, and JetBrains Mono. Cinematic 16:9 page geometry. Bone (#f0e8d0) typography on near-void (#050507). Rose (#cc90a8) accent only. Built with ReportLab.